UNIVERSITY OF PERADENIYA

DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE

END SEMESTER EXAMINATION - SEMESTER I (2022/2023)

CSC1041 - PROGRAMMING LABORATORY I

Answer **ALL** Questions | Time Allowed: **Three Hours**

PART A - (C PROGRAMMING)

Use the vi editor and save your C program as S21XXXQ1.c where S21XXX stands for your registration number. Extra marks will be awarded for using good programming practices.

1. **[50 Marks]**
   You are required to create a C program to manage book rentals from a bookstore called "Book Heaven". The program should collect user data, allow users to rent books, calculate rental costs including potential discounts, and generate rental receipts. The program should also have the capability to view current rentals and clear all rental records. A recurring Menu should be implemented as given below.

Example Output:

```
Welcome to Book Heaven!

1. Rent books
2. Print book price list
3. View current rentals
4. Clear rental records
5. Exit

Please enter your choice:
```

a. When Option 1 (Rent books) is selected, the program should ask for user data.

   User data: Customer name, Customer ID, Mobile Number, Rental Period (in days)

   Book Rental:

   - A customer can rent a maximum of 5 different books at a time.
   - After renting a book, the program should ask if the customer wants to rent another book. If the user inputs an invalid book number, the rental process ends.
   - The available books and their **daily rental prices** are:
     - "The Great Gatsby" - Rs. 50.00
     - "1984" - Rs. 60.00
     - "To Kill a Mockingbird" - Rs. 40.00
     - "Moby-Dick" - Rs. 30.00
     - "War and Peace" - Rs. 70.00
     - "Pride and Prejudice" - Rs. 50.00

Discount Calculation:

- If the rental period is more than 7 days, apply a 10% discount on the total rental price.
- If the rental period is more than 14 days, apply a 20% discount on the total rental price.

Receipt Generation:

- After renting books, generate and print a receipt that includes:
  - Customer details
  - Rented books and their rental periods
  - Total rental price
  - Discount applied (if any)
  - Final total price after discount

Example Output:

```
Please enter your choice: 1
Enter customer name: Alex
Enter customer ID: 23456789
Enter mobile number: 0768543219
Enter rental period (in days): 9

Available books:
1. The Great Gatsby - Rs. 50.00 per day
2. 1984 - Rs. 60.00 per day
3. To Kill a Mockingbird - Rs. 40.00 per day
4. Moby-Dick - Rs. 30.00 per day
5. War and Peace - Rs. 70.00 per day
6. Pride and Prejudice - Rs. 50.00 per day

Enter book 1 (1-6): 3
Do you want to rent another book? (y/n): y
Enter book 2 (1-6): 6
Do you want to rent another book? (y/n): y
Enter book number (1-6): 7
Invalid book number. Rental process ended.

Books rented successfully! Here is your receipt:

Book Haven
Customer Name: Alex
Customer ID: 23456789
Mobile: 0768543219
Rental Period: 9 days

Rented Books:
To Kill a Mockingbird - Rs. 40.00 per day - Total: Rs. 360.00
Pride and Prejudice - Rs. 50.00 per day - Total: Rs. 450.00

Total Rental Price: Rs. 810.00
Discount Applied: 10% (Rs. 81.00)
Final Total Price: Rs. 729.00
Thank you for renting with us!
```

b. When Option 2 (Print book price list) is selected, the details of all the rental books should display as follows.

Example Output:

```
Please enter your choice: 2
Book Price List:
1. The Great Gatsby - Rs. 50.00 per day
2. 1984 - Rs. 60.00 per day
3. To Kill a Mockingbird - Rs. 40.00 per day
4. Moby-Dick - Rs. 30.00 per day
5. War and Peace - Rs. 70.00 per day
6. Pride and Prejudice - Rs. 50.00 per day
```

c. When Option 3 (View current rentals) selected, all current rentals with customer details and rented books should display as follows.

Example Output:

```
Please enter your choice: 3
Current Rentals:

Customer Name: Alex
Customer ID: 23456789
Mobile: 0768543219
Rental Period: 9 days
Rented Books:
To Kill a Mockingbird - Rs. 40.00 per day
Pride and Prejudice - Rs. 50.00 per day
```

d. When Option 4 (Clear rental records), Clear all existing rental records, effectively resetting the rental system.

Example Output:

```
Please enter your choice: 4
All rental records have been cleared.

Welcome to Book Haven!
1. Rent books
2. Print book price list
3. View current rentals
4. Clear rental records
5. Exit
Please enter your choice: 3
No current rentals.
```

PART B - (PYTHON PROGRAMMING)

Create a folder named CSC1041_End_S21XXX in your H drive. Save your Python programs inside the same folder as S21XXXQ2.py, S21XXXQ3.py and S21XXXQ4.py where S21XXX is your registration number.

1. **[24 Marks]**
   a. Write a Python function to pass a filename and a String context as parameters and open the file. Then write the context to the file and close it. Please note that an error message should display if the file is already existing.

   b. Using the above function written in (a) open a new file named *context_1.txt* and write the following sentence to it.

Is Machine learning a very advanced level concept?

c. Then open a new file named *context_2.txt* and write the following sentence.

Madam it is amazing!

d. Write a function to pass a sentence as a parameter and convert all letters in that sentence to lowercase.

e. Write a Python function that takes two file names, merges their contents, converts all letters to lowercase and writes it into a third file named *merged_file.txt*. Ensure that the first file's contents appear before the second file's contents in the merged file. Call this function using files *context_1.txt* and *context_2.txt*.

f. Count the total number of words in the *merged_file.txt* and print the count.

g. Write a Python function that checks if a given string contains palindrome words (a word that reads the same backwards as forward, e.g., madam or racecar) and print palindrome words if there are any.

h. Test the above function in part (g) using the content in *merged_file.txt*. (You need a separate function to read the content from a file.)

2. [08 Marks]

a. Write a Python program that takes a sentence as a user input and,

   i. Convert all letters in the sentence into uppercase and print them.

   ii. Count and print the number of vowels in the sentence.

Give the following sentence as a user input.

```
I like ChatGPT.
```

The expected outcome is as follows,

```
Enter a sentence: I like ChatGPT.
Uppercase sentence: I LIKE CHATGPT.
Number of vowels: 4
```

b. Print the following sentence as it is.

"The cat's tail: a
mystery in motion."

3. [18 Marks]

The following dictionary shows products and their prices in an inventory, perform the following tasks using the given dictionary,

```
inventory = {
  "Laptop": 999.99,
  "Smartphone": 499.90,
  "Tablet": 299.45,
  "Headphones": 89.99,
```

```
        "Smartwatch": 199.65,
        "Monitor": 159.99,
        "Keyboard": 49.50,
        "Mouse": 29.99,
        "Printer": 119.75,
        "Desk Lamp": 39.99
}
```

a. Create a Python dictionary named inventory and store the above data.

b. Remove the *Keyboard* from the inventory. (Both the product and the price.)

c. Add the following new item to the inventory.
```
"Writing Pad": 122.45
```

d. Print all the items in the updated inventory.

e. Write a Python function to calculate and return the total cost of all products in the inventory.

f. Write a Python function to calculate and return the average price of the products in the inventory.

g. Write a Python function to categorize the products into "Expensive" (price > $500) and "Affordable" (price <= $500) categories.

The expected outcome is as follows,

```
Updated Inventory items:
Laptop: $999.99
Smartphone: $499.90
Tablet: $299.45
Headphones: $89.99
Smartwatch: $199.65
Monitor: $159.99
Mouse: $29.99
Printer: $119.75
Desk Lamp: $39.99
Writing Pad: $122.45

Total Cost of Inventory: $2561.10
Average Price of Inventory: $256.11

Expensive Products:
Laptop: $999.99

Affordable Products:
Smartphone: $499.90
Tablet: $299.45
Headphones: $89.99
Smartwatch: $199.65
Monitor: $159.99
Mouse: $29.99
Printer: $119.75
Desk Lamp: $39.99
Writing Pad: $122.45
```

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